

Computational Logic

1- Semantics of Propositional Logic

(lab classes 1 and 2)

⇒ 1.1 For each of the following sentences, state whether it is a proposition. If so, state whether it is true or false.

- | | |
|---|--|
| (a) Porto is the capital of Portugal. | (e) $1 + 2 + 3 = 6$ |
| (b) In 1910 the Portuguese Republic was proclaimed. | (f) 7 is an even number. |
| (c) Shall I compare thee to a summer's day. | (g) $x + y$ is an even number. |
| (d) $1 + 2 + 3$ | (h) There is an x such that $7 + x$ is an even number. |

1.2. Let p and q be the propositions "I play the lottery" and "I win the *jackpot*", respectively. Express each of the following compound propositions as a sentence in natural language (English or Portuguese):

- | | |
|--------------------------|---------------------------------|
| (a) $\neg p$ | (d) $p \rightarrow q$ |
| (b) $p \wedge q$ | (e) $q \rightarrow p$ |
| (c) $\neg p \vee \neg q$ | (f) $\neg p \rightarrow \neg q$ |

⇒ 1.3 Considering the priority conventions of logical operations given in the course ($\neg < \wedge < \vee < \rightarrow$ and $\rightarrow$ right associative), for each of the following formulas draw its syntactic tree and indicate its subformulas:

- (a) $p \wedge \neg q \rightarrow \neg p$;
- (b) $p \wedge (\neg q \rightarrow \neg p)$;
- (c) $\neg(s \rightarrow (\neg(p \rightarrow (q \vee \neg s))))$;
- (d) $p \rightarrow (\neg q \vee (q \rightarrow p))$;
- (e) $(p \rightarrow q) \wedge (\neg r \rightarrow (q \vee (\neg p \wedge r)))$.

⇒ 1.4 Construct truth tables for each of the following propositions.

$$(a) p \rightarrow p$$

$$(b) \neg q \wedge (p \rightarrow q)$$

$$(c) (p \wedge q) \rightarrow \neg(p \vee q)$$

$$(d) \neg(p \vee q) \rightarrow (\neg p \wedge \neg q)$$

$\Rightarrow$ 1.5 When $v(p) = \mathbf{T}$ and $v(q) = \mathbf{F}$ determine if $\models_v \phi$, where ϕ is:

$$(a) (p \vee q) \rightarrow (p \wedge q);$$

$$(b) (p \vee \neg q) \wedge (\neg p \wedge q);$$

$$(c) (p \rightarrow q) \wedge (p \rightarrow \neg q);$$

$$(d) p \rightarrow (q \rightarrow (p \vee q));$$

$$(e) ((p \rightarrow q) \rightarrow p) \rightarrow p;$$

1.6. Using truth tables or the semantic equivalences given in class (boolean algebra), show the following equivalences between propositions.

$$(a) \neg(p \rightarrow q) \Leftrightarrow p \wedge \neg q$$

$$(b) \neg p \rightarrow q \Leftrightarrow \neg q \rightarrow p$$

$$(c) p \rightarrow (q \rightarrow r) \Leftrightarrow q \rightarrow (p \rightarrow r)$$

$$(d) (p \rightarrow q) \rightarrow p \Leftrightarrow p$$

$$(e) (p \vee q) \rightarrow r \Leftrightarrow (p \rightarrow r) \wedge (q \rightarrow r)$$

1.7. Assigning truth values to p , q , and r investigate how many of the disjunctions $p \vee \neg q$, $\neg p \vee q$, $q \vee r$, $q \vee \neg r$, $\neg q \vee \neg r$ can be simultaneously true.

$\Rightarrow$ 1.8 Which of the following statements are valid? Justify.

$$(a) (p \rightarrow q) \wedge (p \rightarrow r) \Leftrightarrow (p \rightarrow (q \wedge r))$$

$$(b) (p \rightarrow r) \wedge (q \rightarrow r) \Leftrightarrow ((p \wedge q) \rightarrow r)$$

$$(c) (p \wedge q) \vee r \Leftrightarrow (p \vee r) \wedge (q \vee r)$$

1.9. For the following formulas indicate, **justifying**, if they are tautologies, satisfiable or contradictions. You can start by simplifying the formulas using semantic equivalences. For each of the formulas that you think is not a tautology or a contradiction, indicate an assignment that makes the formula true and another that makes the formula false.

$$(a) (p \rightarrow (q \vee r)) \vee (r \rightarrow \neg p);$$

$$(b) p \rightarrow (q \rightarrow \neg(\neg q \rightarrow \neg p));$$

$$(c) (p \rightarrow q) \rightarrow (\neg p \rightarrow \neg q);$$

$$(d) p \rightarrow (q \rightarrow (p \vee q));$$

$$(e) (q \rightarrow r) \rightarrow ((\neg q \rightarrow \neg p) \rightarrow (p \rightarrow r));$$

$$(f) (p \vee q) \rightarrow (q \rightarrow (p \wedge q));$$

$$(g) (p \rightarrow q) \wedge \neg((q \rightarrow s) \rightarrow (p \rightarrow s));$$

$$(h) (p \rightarrow q) \wedge (\neg r \rightarrow (q \vee (\neg p \wedge r)));$$

$$(i) (p \vee q) \rightarrow (p \wedge q)$$

$$(j) (p \vee \neg q) \wedge (\neg p \wedge q)$$

$$(k) ((p \vee q) \wedge r) \leftrightarrow ((p \wedge r) \vee (q \wedge r))$$

$\Rightarrow$ **1.10** ¹ Imagine an island where some of its inhabitants, called “knights”, always tell the truth, and others called “knaves” always lie.

We assume that any inhabitant of this island falls into one or the other class.

- (a) Two inhabitants, **A** and **B** are in a garden. **A** makes the following statement: “At least one of us is a knave.”
What are **A** and **B**?
- (b) Two inhabitants, **A** and **B** are in a garden. Suppose **A** says, “I am a knave, or **B** is a knight.”
What are **A** and **B**?
- (c) Two inhabitants, **A** and **B**, are in a garden.
Suppose **A** says, “I am a knave, or two plus two equals five.” What can we conclude?
- (d) Two inhabitants of the island, **A** and **B**, were together in a garden.
Suppose **A** says, “I am a knave, but **B** is not.”
What are **A** and **B**?
- (e) Three townspeople, **A**, **B**, and **C** come together. **A** and **B** make the following statements:
A: We are all knaves.
B: Exactly one of us is a knight.
What are **A**, **B**, and **C**?
- (f) Suppose that in the previous problem, **A** and **B** say the following:
A: We are all knaves.
B: Exactly one of us is a knave.
Can we determine what **B** is? Can we determine what **C** is?
- (g) We have three inhabitants, **A**, **B**, and **C** (knights or knaves). **A** says: “**B** and **C** are of the same kind.” Someone then asks **C**: “**A** and **B** are of the same kind?”
What did **C** say?
- (h) Once again we are faced with three inhabitants, **A**, **B**, and **C**. Two of them are of the same type (knights or knaves). **A** and **B** make the following statements:
A: **B** is a knave.
B: **A** and **C** are of the same type.
What is **C**?
- (i) Three of the island’s inhabitants, **A**, **B**, and **C**, were together in a garden. A stranger who was passing by asked **A**, “Are you a knight or a knave?” **A** answered, but in such a way that the stranger could not understand the answer. The foreigner then asked **B**, “What did **A** say?” **B** replied: “**A** said he was a knave.” At that point **C** exclaimed: “Don’t believe **B**, he’s lying!”
What are **B** and **C**?
- (j) On that island there is a fork in the road, one of the paths leading to the nearest city. If a foreigner wants to know which is the right path, what question should he ask an inhabitant at the fork, in such a way that he can answer with a simple “yes” or “no”. **Suggestion**: The question should relate (by equivalence) what you want to know with the determination of the class of the inhabitant.

¹Excerpt from *What is the name of this book?*, R. Smullyan

- (k) **An adventure of mine** This is an unusual problem, and besides it is inspired by a real case. Once when on the island of knights and knaves, I found two inhabitants resting under a tree. I approached and asked one of them: "Is either of you a knight?". He answered, and with that I learned the answer to my question.

Was the person I questioned a knight or a knave? And the other? I can assure you that I gave the necessary information to solve the problem.

- (l) Suppose you visited the island of knights and knaves and found two of its inhabitants lazily resting in the sun. You ask one of them if the other is a knight and you receive the answer (yes or no). Then you ask the other if the first is a knight and you receive the respective answer (yes or no). Are the two answers necessarily the same?

1.11. Three individuals, **A**, **B**, and **C**, suspected of a crime, make the following statements, respectively:

A: **B** is guilty, but **C** is innocent;

B: If **A** is guilty, then **C** is guilty;

C: I am innocent, but one of the other two is guilty.

- Are the three statements compatible?
- Are any of the statements a consequence of the other two?
- Assuming all three defendants are innocent, who lied?
- Assuming that they all told the truth, who is innocent and who is guilty?
- Assuming that the innocent told the truth and the guilty lied, who is innocent and who is guilty?

⇒ 1.12 Imagine the following rules for choosing optional subjects for the next academic year:

R1: If you choose *Computer Graphics*, then you must also choose *Numerical Programming* or *Algebra*.

R2: Choose *Algebra* if and only if you choose *Computer Graphics* or *Numerical Programming*.

R3: You cannot choose all three, but you must choose at least one of the three subjects.

R4: If you do not choose *Algebra*, then choose the other two.

- Is it possible to satisfy this set of rules?
- Is rule R4 really necessary? What about rule R1?
- Give a simple rule that is equivalent to this (messy) set of four rules.

1.13. Justify the truth or falsity of each of the following statements where p, q, r and s are propositional variables.

- $\models (p \wedge \neg p) \rightarrow (p \wedge \neg r)$.
- $\models (q \vee \neg r) \rightarrow (q \wedge \neg q)$.
- $\{(p \rightarrow q) \vee r, ((p \rightarrow q) \vee r) \rightarrow \neg r, (p \rightarrow q) \rightarrow (q \rightarrow r)\} \models \neg q$.
- $\{(p \vee q) \rightarrow r\} \models (p \rightarrow r) \vee (q \rightarrow r)$.

⇒ 1.14 Without building the truth table, justify the truth or falsity of each of the following statements where p, q, r and s are propositional variables.

- $\models (p \rightarrow (((q \rightarrow p) \rightarrow p) \rightarrow r))$.

- (b) $\models (p \rightarrow (q \rightarrow s)) \rightarrow (q \rightarrow (p \rightarrow s))$.
 (c) $\models (p \rightarrow (q \rightarrow r)) \rightarrow ((p \rightarrow q) \rightarrow (p \rightarrow r))$.
 (d) $\models ((p \wedge q) \rightarrow (s \vee t)) \rightarrow ((p \rightarrow s) \vee (q \rightarrow t))$.

1.15. Justify the truth or falsity of each of the following statements, where Γ and Σ represent sets of formulas and $\phi, \psi, \theta, \gamma$ represent formulas of propositional logic. If the statement is false, you must give a counterexample indicating particular values of the sets or formulas for which the statement is false. If the statement is true you must justify using the definitions.

- (a) $\models (\theta \vee \phi) \rightarrow (\neg\theta \rightarrow \phi)$.
 (b) $\{\phi \rightarrow \psi, \psi \rightarrow \phi\} \models (\phi \vee \psi) \rightarrow (\phi \wedge \psi)$.
 (c) $\{\psi \rightarrow (\neg\phi \vee \gamma), \neg\gamma\} \models \neg\phi \rightarrow \neg\psi$.
 (d) If $\phi, \psi \models \theta \rightarrow \gamma$ and $\phi \models \theta$ then $\phi, \psi \models \gamma$.

$\Rightarrow$ **1.16** Justify the truth or falsity of each of the following statements, where Γ and Σ represent sets of formulas and $\phi, \psi, \theta, \gamma$ represent formulas of propositional logic. If the statement is false, you must give a counterexample indicating particular values of the sets or formulas for which the statement is false. If the statement is true you must justify using the definitions.

- (a) If $\Gamma \models \theta$ and $\Gamma \subseteq \Sigma$, then $\Sigma \models \theta$.
 (b) If $\Sigma \models \theta$ and $\Gamma \subseteq \Sigma$, then $\Gamma \models \theta$.
 (c) If $\Gamma \models \theta$ and $\Sigma \models \theta$, then $\Gamma \cap \Sigma \models \theta$.
 (d) If $\Gamma \models \theta$ and $\Sigma \subseteq \Gamma$ then $\Sigma \cap \Gamma \models \theta$.
 (e) If $\Sigma \models \theta$ then $\Sigma \cap \Gamma \models \theta$.
 (f) If $\Sigma \models \theta$ and $\Sigma \subseteq \Gamma$ then $\Sigma \cup \Gamma \models \theta$.

$\Rightarrow$ **1.17** For each of the truth functions (a), (b) and (c), construct a formula in disjunctive normal form and a formula in conjunctive normal form:

p	q	r	(a)	(b)	(c)
T	T	T	T	F	T
T	T	F	T	T	F
T	F	T	T	F	T
T	F	F	T	T	F
F	T	T	F	T	T
F	T	F	T	T	F
F	F	T	T	F	T
F	F	F	T	F	F

1.18. For each of the following formulas determine an equivalent formula in disjunctive normal form and another in conjunctive normal form.

- (a) $\neg(p \rightarrow (\neg(q \wedge (\neg p \rightarrow q))))$;
 (b) $p \rightarrow (\neg q \rightarrow \neg(p \rightarrow q))$;
 (c) $\neg(p \rightarrow (\neg(q \wedge (\neg p \rightarrow q))))$;
 (d) $(p \rightarrow q) \rightarrow (\neg p \rightarrow \neg q)$;

- (e) $((p \rightarrow q) \rightarrow p) \rightarrow p$;
- (f) $p \rightarrow \neg(q \rightarrow \neg(p \rightarrow q))$;
- (g) $(p \rightarrow q) \wedge (\neg r \rightarrow (q \vee (\neg p \wedge r)))$;
- (h) $(p \rightarrow q) \wedge (\neg q \rightarrow (p \vee \neg q))$;
- (i) $p \rightarrow (q \rightarrow \neg(\neg q \rightarrow \neg p))$;
- (j) $p \rightarrow (q \rightarrow (p \wedge q))$;
- (k) $(p \rightarrow q) \rightarrow (\neg p \rightarrow \neg q)$;
- (l) $((p \vee q) \rightarrow (r \wedge s)) \rightarrow (p \vee \neg r)$;
- (m) $(p \rightarrow q) \wedge \neg((q \rightarrow s) \rightarrow (p \rightarrow s))$;

$\Rightarrow$ 1.19 Sets of connectives

- (a) Show that the logical connectives $\{\neg, \vee\}$ are a universal set. That is, propositions of the form $p \rightarrow q$, $p \leftrightarrow q$, and $p \wedge q$ can be written using only the connectives $\neg$ and $\vee$.
- (b) Define a new connective $\overline{\wedge}$ as follows:

p	q	$p \overline{\wedge} q$
T	T	F
T	F	T
F	T	T
F	F	T

Show that the propositions $\neg p$, $p \wedge q$, and $p \vee q$ can be written using only the connective $\overline{\wedge}$. Conclude that $\overline{\wedge}$ is universal.

1.20. Write a program to manipulate propositional logic formulas with the basic connectives: negation, disjunction, conjunction and implication.

- (a) Given a formula must allow:
 - given a valuation of the variables, calculate the valuation of the formula
 - calculate the truth table of the formula
 - determine whether the formula is satisfiable, contradiction or tautology
 - obtain a formula in equivalent conjunctive normal form
 - obtain a formula in equivalent disjunctive normal form
- (b) Given two formulas, determine whether one is a semantic consequence of the other, concluding whether or not they are equivalent.

$\Rightarrow$ 1.21 Consider the satisfiability algorithm for Horn's formulas given in the course.

- (a) Describe the algorithm in pseudo-code and implements it.
- (b) Justify the correctness of the algorithm, that is, it assigns the value true if and only if Horn's formula is satisfiable.
- (c) Apply the algorithm to the following formulas:
 - $(\neg p \vee \neg q) \wedge (\neg q \vee r) \wedge q$
 - $p \wedge (\neg p \vee q) \wedge (\neg q \vee p)$

$$\begin{aligned}
& - \neg p \wedge (\neg p \vee q) \wedge \neg q \\
& - p \wedge (\neg p \vee q) \wedge \neg r
\end{aligned}$$

⇒ 1.22 Apply the David-Putman (DLL) algorithm to the following sets of clauses.

- (a) $\{p \vee q \vee r, \neg p \vee \neg q \vee \neg r, p \vee \neg q \vee \neg r, p \vee q \vee \neg r, \neg p \vee q, \neg p \vee r, p \vee \neg q \vee r\}$
- (b) $\{\neg p \vee q \vee r, p \vee q \vee r, p \vee q \vee \neg r, p \vee \neg q \vee r, p \vee \neg q \vee \neg r\}$
- (c) $\{p \vee \neg q \vee \neg r, \neg p \vee q \vee r, \neg p \vee q \vee \neg r, \neg p \vee q, \neg p \vee r\}$
- (d) $\{p \vee \neg q \vee r, \neg p \vee q \vee r, p \vee \neg q \vee \neg r, \neg p \vee q \vee \neg r, \neg p \vee \neg q \vee \neg r\}$
- (e) $\{p \vee \neg q \vee \neg r, \neg p \vee q \vee r, \neg p \vee \neg q \vee \neg r, \neg p \vee \neg q \vee r, \neg p \vee q \vee \neg r, p \vee q \vee \neg r\}$

1.23. Implement the David-Putman algorithm for clause satisfiability.

Laws of semantic equivalence

$\phi \wedge \psi$	$\Leftrightarrow$	$\psi \wedge \phi$	(commutativity)
$\phi \vee \psi$	$\Leftrightarrow$	$\psi \vee \phi$	(commutativity)
$\neg(\phi \wedge \psi)$	$\Leftrightarrow$	$(\neg\phi \vee \neg\psi)$	(DeMorgan's Law)
$\neg(\phi \vee \psi)$	$\Leftrightarrow$	$(\neg\phi \wedge \neg\psi)$	(DeMorgan's Law)
$(\phi \wedge \psi) \wedge \theta$	$\Leftrightarrow$	$\phi \wedge (\psi \wedge \theta)$	(associativity)
$(\phi \vee \psi) \vee \theta$	$\Leftrightarrow$	$\phi \vee (\psi \vee \theta)$	(associativity)
$(\phi \vee \phi)$	$\Leftrightarrow$	ϕ	(idempotence)
$(\phi \wedge \phi)$	$\Leftrightarrow$	ϕ	(idempotence)
$(\phi \wedge \psi) \vee \theta$	$\Leftrightarrow$	$(\phi \vee \theta) \wedge (\psi \vee \theta)$	(distributiveness)
$(\phi \vee \psi) \wedge \theta$	$\Leftrightarrow$	$(\phi \wedge \theta) \vee (\psi \wedge \theta)$	(distributivity)
$\neg\neg\phi$	$\Leftrightarrow$	ϕ	(Double negation)
$\phi \rightarrow \psi$	$\Leftrightarrow$	$\neg\phi \vee \psi$	